

Research Document on NOVA Buildathon 2026

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Mind Wandering Detection

Processing Data from COG-BCI Dataset

The **COG-BCI database** [1] is a multi-session, multi-task EEG dataset designed for research on **passive brain-computer interfaces** (pBCI). Each subject performed the same battery of cognitive tasks across three sessions while **63 EEG channels** were recorded at **500 Hz** with a **common-average reference**. For every condition, the dataset also provides trial-level **behavioral logs**, which makes it possible to align neural activity with overt performance.

Dataset Structure

The local copy follows a BIDS-like layout, organized per subject (sub-XX) and session (ses-S1–ses-S3):

- eeg/ – raw EEGLAB files (.set header + .fdt binary data) for each condition;
- behavioral/ – trial-level behavioral logs (.mat) mirroring the same conditions;
- chanlocs/ – channel names and 3D coordinates.

Each session contains the same task battery: the N-back task (0/1/2-back), the Flanker task, the Multi-Attribute Task Battery (MATB), the Psychomotor Vigilance Task (PVT), and resting-state blocks (eyes open / eyes closed) at the start and end of the session.

Psychomotor Vigilance Task

The **Psychomotor Vigilance Task** (PVT) is a classic **sustained-attention** paradigm. Participants watch a screen and press a button as soon as a stimulus appears. Stimuli are presented at **pseudo-random inter-stimulus intervals** of 2–10 s, so the task cannot be solved by rhythm or anticipation; performance therefore tracks the current level of vigilance.

The primary outcome is the **reaction time** (RT) on each trial. Slow responses and **lapses** are the hallmark of reduced vigilance, which is observed under sleepiness, fatigue, and **mind wandering**. When the mind wanders, attention is diverted away from the stimulus, so the participant reacts late or not at all – precisely the lapse signature that the PVT measures. PVT performance is therefore a standard **behavioral proxy** for mind-wandering episodes, and, combined with the simultaneously recorded EEG, it allows searching for the neural correlates of attentional drift.

The PVT EEG is stored in **EEGLAB format**: a .set header file together with a .fdt binary file that holds the raw samples. Inspecting the MATLAB struct of PVT.set gives the following picture:

```
EEG =  
  filename: 'PVT.set'   datfile: 'PVT.fdt'  
  nbchan:   63         trials: 1  
  pnts:    306060      srate:  500  
  xmin:     0         xmax:   612.118  
  data:    [63×306060 single]  
  ref:     'common'  
  event:    [272×1 struct]
```

The various fields have the following meaning:

- nbchan: 63, srate: 500 – the recording contains **63 EEG channels** sampled at **500 Hz**.
- trials: 1, pnts: 306060, xmax: 612.118 – a **single continuous recording** of 306060 sample points, i.e. roughly **612 seconds (~10.2 minutes)**, that is **not** segmented into epochs.
- data: [63×306060 single] – the raw amplitude values (in microvolts, single-precision floats) laid out as **channels × time points**.

- ref: 'common' – the signal has already been **average-referenced**.
- chanlocs: [1×63 struct] – the channel locations, matching chanlocs/get_chanlocs.txt.
- event: [272×1 struct] – event markers for the 90 PVT trials (stimulus onsets and responses) together with session start/end markers; these events make it possible to **segment the continuous signal into trials**.
- datfile: 'PVT.fdt' – the companion binary file containing the actual samples.

The behavioral log PVT.mat describes the same 90 trials at the performance level:

Field	Value	Meaning
reaction_times	1×90 double	Reaction time (s) of each of the 90 trials
error_trial	90×1 double	Binary flag marking erroneous trials (anticipations / misses)
reaction_time_error_trial	1×90 double	Reaction time for error trials; large negative sentinel values where no valid RT exists
isi_time	90×1 double	Inter-stimulus interval preceding each trial
simin / simax	2 / 10	Design bounds of the ISI in seconds
ntrials	90	Total number of trials
triggers	5×2 table	Counts of the trigger codes used in the task

Taken together, PVT.set / PVT.fdt provide the continuous neural signal while PVT.mat provides the per-trial behavioral ground truth; aligning the two (via event and isi_time) yields trials labeled by reaction time, which can be thresholded into attentive vs. mind-wandering episodes.

References

- [1] M. F. Hinss, E. S. Jahanpour, B. Somon, L. Pluchon, F. Dehais, and R. N. Roy, *COG-BCI database: A multi-session and multi-task EEG cognitive dataset for passive brain-computer interfaces*. (Jul. 2022). Zenodo. doi: 10.5281/zenodo.6874129.